

Version 2.0 of the hglm package

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New features in the hglm package

Available on CRAN soon (available now on R Forge).

- Spatial modelling (eg `rand.family = CAR(D = nbr)`)
- HL11 correction for non-normal outcomes (`method = "HL11"`)
- Several variance components from different distributions (eg `rand.family = list(Gamma(), gaussian())`)
- Possible to have a linear predictor for the variance component of the random effect (eg `X.rand.disp = X`)

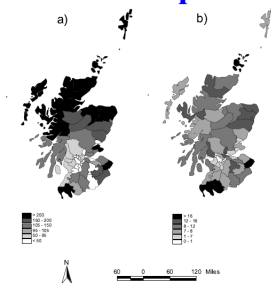
A spatial GLMM with random effects being distributed as the Gaussian Markov Random Field (GMRF) is presented as follows

$$E(y_s|u_s) = \mu_s, \quad g(\mu_s) = \eta_s = \mathbf{X}_s^T \boldsymbol{\beta} + \mathbf{Z}_s u$$

$$u \sim N(\mathbf{0}, \Sigma = \tau(\mathbf{I} - \rho D)^{-1})$$

where s represents a location given by some coordinates, $\boldsymbol{\beta}$ is a vector of fixed effects, $\mathbf{X}_s$ is the vector observed covariates at location s and u being the location specific random effects. The D matrix is a known “neighbourhood matrix”. The two parameters τ and ρ are estimated.

Dataset: Scottish lip cancer data



- a) $SMR_i = 100Y_i/E_i$, **standardized mortality ratio** for lip cancer in $I = 56$ districts, 1975–1980
- b) one covariate, x_i = percentage of the population engaged in **agriculture, fishing or forestry (AFF)**

```
library(hglm) #Requires version 2.0
data(cancer)
logE <- log(E); X11 <- model.matrix(~Paff)
m41 <- hglm(X = X11, y = 0, Z = diag(length(0)),
  family = poisson(), rand.family = CAR(D = nbr),
  offset = logE, conv = 1e-9, maxit = 200, fix.disp = 1)
```

Linear predictor and different distributions for random effects

```
hglm(X = X, y = y, Z = Z,  
      X.rand.disp = X2)
```

```
hglm(X = X, y = y, Z = cbind(Z1, Z2),  
      rand.family = list(CAR(D = D), gaussian()),  
      RandC = c(100, 20))
```

HL11 correction - a simulation study

Fitting a Poisson GLMM with iid random effects u of length equal to the number of observations

```
hglm(X = model.matrix(y~1), y = y, Z = diag(length(y)),  
     family = poisson(), fix.disp = 1,  
     method = "HL11")
```

```
[1] "EQL results"  
      Intercept      Variance component      Converged  
Min.      :-11.91495      Min.      : 0.3614      Min.      :0.00  
1st Qu.:  0.16231      1st Qu.:  0.6048      1st Qu.:1.00  
Median   :  0.22143      Median   :  0.7297      Median   :1.00  
Mean     : -0.01959      Mean     :  4.8199      Mean     :0.98  
3rd Qu.:  0.28766      3rd Qu.:  0.8887      3rd Qu.:1.00  
Max.     :  0.46192      Max.     :205.0796      Max.     :1.00  
[1] "HL11 results"  
      Intercept      Variance component      Converged  
Min.      :-0.388793      Min.      :0.4561      Min.      :1  
1st Qu.:  0.001421      1st Qu.:0.7739      1st Qu.:1  
Median   :  0.080694      Median   :0.9479      Median   :1  
Mean     :  0.069348      Mean     :0.9530      Mean     :1  
3rd Qu.:  0.154920      3rd Qu.:1.0901      3rd Qu.:1  
Max.     :  0.349771      Max.     :1.6615      Max.     :1
```